COMPSCI 589 - Final Project: Spring 2026

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Extra Credit Declarations

[Extra Credit Question #1] (10 Points): We analyzed all four datasets using an additional algorithm. For Dataset 1, we evaluated three algorithms (k-NN, Random Forest, and Neural Networks) rather than the required two.

1 Experiments and Analyses

1.1 Dataset 1.1: The Hand-Written Digits Recognition Dataset

Task Assigned To: Marwan Hegab.
Implementations Used: Peter's implementations of the k-NN and Random Forest algorithms. Additionally, Marwan's implementation of the Neural Networks.

Algorithm Selection Rationale: k-NN and Neural Networks were primary choices for this dataset because they traditionally preform relatively well at image recognition and capturing relationships in pixel data. Additionally, Random Forest serves as a strong, non-linear ensemble baseline to compare against.

- Each instance of the dataset represents an 8×8 grayscale image resulting in 64 numerical attributes. We globally normalized all pixel intensities by dividing them by 16.0 (the maximum pixel value) prior to training to ensure scale consistency across all three algorithms.

- The performance of each of the three algorithms was evaluated using Stratified 10-Fold Cross-Validation, measuring both Accuracy and F1-Score (Weighted).

1.1.1 Hyperparameter Tuning

At least 6 different hyperparameter settings were used for each algorithm. The results averaged across the 10 folds are shown below:

1. k-Nearest Neighbors (k-NN):

Hyperparameter (k)	Accuracy	F1-Score
$k = 1$	**0.9894**	**0.9894**
$k = 3$	0.9894	0.9893
$k = 5$	0.9861	0.9860
$k = 7$	0.9861	0.9861
$k = 9$	0.9833	0.9833
$k = 15$	0.9805	0.9805

The optimal hyperparameter for k-NN is $k = 1$.

2. Random Forest:

Hyperparameter (# of Trees)	Accuracy	F1-Score
$trees = 1$	0.8046	0.8036
$trees = 5$	0.9305	0.9302
$trees = 10$	0.9555	0.9556
$trees = 15$	0.9616	0.9616
$trees = 20$	0.9694	0.9695
$trees = 30$	**0.9722**	**0.9721**

The optimal hyperparameter for Random Forest is $trees = 30$.

3. Neural Network (Base: Epochs=250, Batch=32):

Hyperparameters (Architecture, Alpha, Lambda)	**Accuracy**	**F1-Score**
Arch=[32, 10], $\alpha = 0.1$, $\lambda = 0.01$ (Baseline)	0.9755	0.9755
Arch=[16, 10], $\alpha = 0.1$, $\lambda = 0.01$ (Shallower)	0.9694	0.9693
Arch=[64, 32, 10], $\alpha = 0.1$, $\lambda = 0.01$ (Deeper)	0.9772	0.9772
Arch=[32, 10], $\alpha = 0.01$, $\lambda = 0.01$ (Low LR)	0.9360	0.9356
Arch=[32, 10], $\alpha = 0.5$, $\lambda = 0.01$ (High LR)	**0.9816**	**0.9816**
Arch=[32, 10], $\alpha = 0.1$, $\lambda = 0.1$ (High Reg)	0.9549	0.9550

The optimal hyperparameters for the Neural Network are Architecture=[32, 10], $\alpha = 0.5$, and $\lambda = 0.01$.

1.1.2 Learning Curves and Analysis

Below are the learning curves for the three evaluated algorithms.

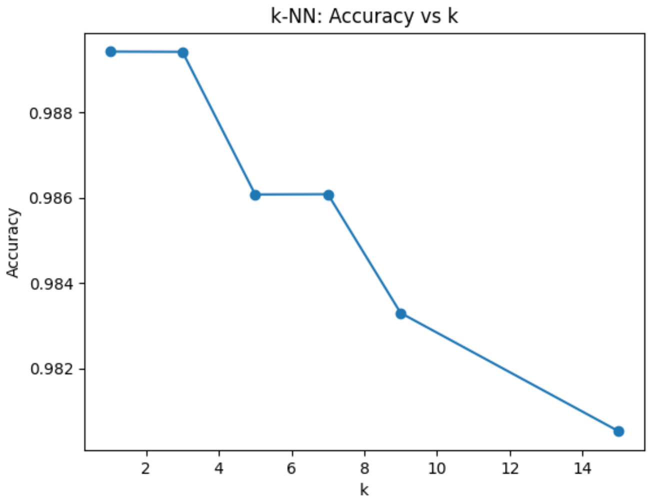

Figure 1: k-NN Learning Curve: Accuracy vs. k

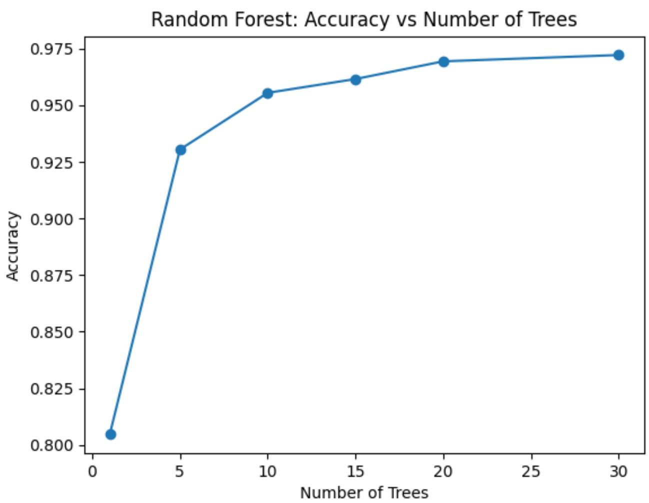

Figure 2: Random Forest Learning Curve: Accuracy vs. Number of Trees

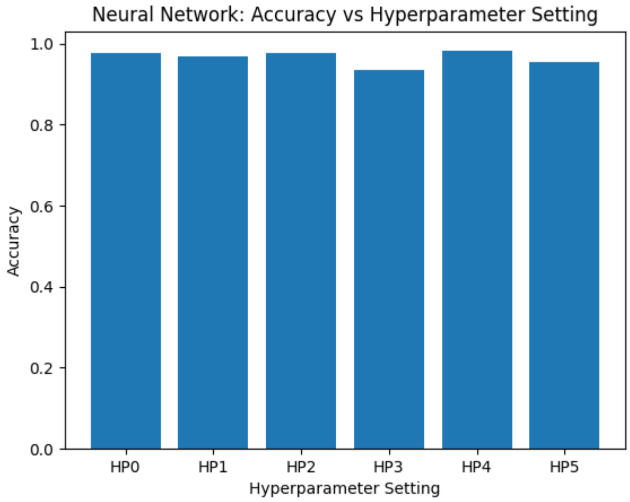

Figure 3: Neural Network Bar Chart: Accuracy vs. Hyperparameter Setting

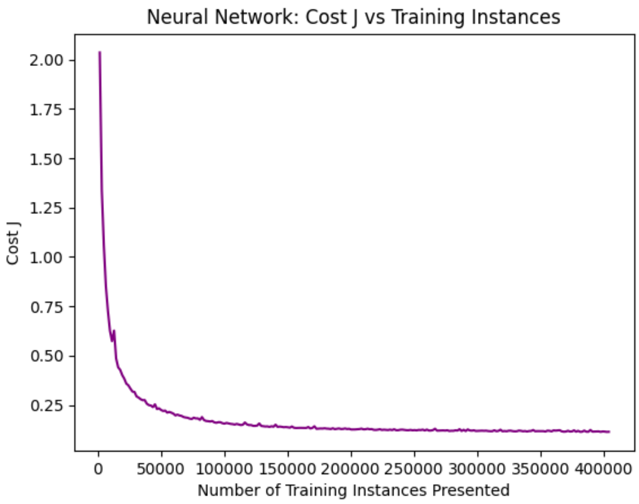

Figure 4: Neural Network Learning Curve: Cost J vs. Training Instances Presented

Analysis and Interpretation: - The k-NN algorithm achieved the highest overall performance on Dataset 1 with an accuracy of 98.94%. In the domain of handwritten digit recognition, image samples of the exact same digit map very closely to each other in Euclidean space because their dark pixels heavily overlap. This results in tight, highly predictive spatial clusters, making k-NN heavily effective, specifically when k is small ($k = 1$ or $k = 3$).

- The Random Forest's performance steadily improved as the number of trees grew to 30, capturing the non-linear relationships between individual pixel intensities and the 10 target classes. The Neural Network performed best with a higher learning rate ($\alpha = 0.5$), suggesting that the digit classification task has a relatively smooth loss landscape that benefits from larger gradient steps. Smaller learning rates (e.g., $\alpha = 0.01$) converged too slowly within the fixed 250 epochs, while stronger regularization ($\lambda = 0.1$) slightly over-constrained the weights, reducing expressiveness on a 10-class problem.

1.1.3 Overall Results Summary

The table below summarizes the best performance achieved by each algorithm on the datasets.

	Dataset 1		Dataset 2		Dataset 3		Dataset 4	
	Accuracy	**F1-Score**	**Accuracy**	**F1-Score**	**Accuracy**	**F1-Score**	**Accuracy**	**F1-Score**
k-NN	0.9894	0.9894						
Random Forest	0.9722	0.9721						
Neural Network	0.9816	0.9816						

Table 1: Overall Performance Summary Table